DGMS Examination 2019 — Question 51 Detailed Solution

Junior Scientific Officer (Geology) | Section F: Geomorphology & Glaciology

QUESTION 51

Which one of the following pairs is NOT correctly matched?

(a) Nunatak : Unglaciated island of bedrock, formerly surrounded by ice

(b) Cirque : Steep-walled semicircular basin in mountain glaciers

(c) Col : Narrow, sharp-edged ridge in mountain glaciers
NOT CORRECTLY MATCHED (CORRECT OPTION)

(d) Fjord : U-shaped glacial trough below sea level

1. Executive Summary & Answer Verification

The correct option to choose is **(c) Col : Narrow, sharp-edged ridge in mountain glaciers**.

In glacial geomorphology:

- A **narrow, sharp-edged, knife-like ridge** separating two adjacent glacial valleys or cirques is universally termed an **Arête** (from the French for "fishbone" or "ridge").
- A **Col** is *not* the sharp ridge itself; rather, it is a **low mountain pass, notch, or saddle** formed along an arête where two opposing cirques erode headward toward each other and breach the intervening ridge crest.

Therefore, statement (c) attributes the diagnostic morphological definition of an *arête* to a *col*, making it NOT correctly matched.

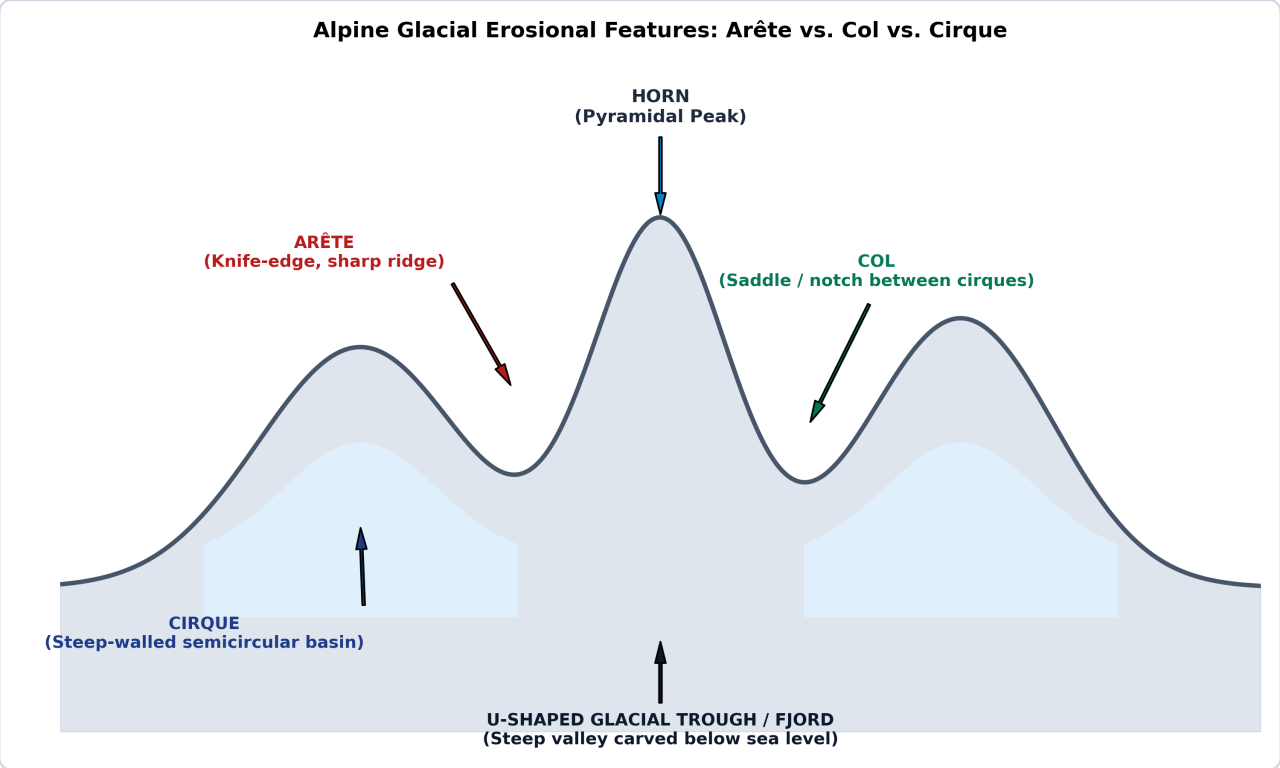


Figure 51.1: *Glacial erosional landforms in alpine mountainous terrains: An **Arête** is the narrow, knife-edge rocky ridge; a **Col** is the saddle-shaped notch or pass breaching the arête between two opposing **Cirques**; a **Horn** is the sharp pyramidal peak formed by 3 or more cirques.*

2. Comprehensive Analysis of Glacial Landforms

2.1 The Col vs. Arête Distinction

During intense alpine glaciation, cirque glaciers erode bedrock by freeze-thaw frost wedging (plucking) and basal abrasion:

- **Arête:** When two parallel alpine valley glaciers or two adjacent cirques erode headward from opposite sides of a divide, the divide is carved into a serrated, razor-sharp, knife-edged rocky ridge called an *arête*.
- **Col:** Continued headward erosion from opposite sides causes the backwalls of two opposing cirques to intersect and break through the intervening arête. This carves a prominent, U- or saddle-shaped depression or pass across the ridge crest known as a *col* (or mountain pass).
- **Horn:** When three or more cirques carve backwalls inward from different sides of a single mountain, they leave a sharp, spire-like pyramidal peak called a glacial *horn* (e.g., the Matterhorn in Switzerland).

2.2 Verification of the Other Three Landforms

1. **(a) Nunatak — Correctly Matched:** A native Inuktitut term for an isolated rock island, peak, or ridge that projects through an ice sheet or ice cap, remaining unglaciated while completely surrounded by flowing glacial ice.
2. **(b) Cirque — Correctly Matched:** A classic amphitheater-shaped, steep-walled, semicircular depression carved into high mountain headwalls by rotational ice slip, frost plucking, and abrasion (also called a *corrie* in Scotland or *cwm* in Wales). Often contains a small glacial tarn lake after ice retreat.
3. **(d) Fjord — Correctly Matched:** A deep, steep-walled, flat-bottomed U-shaped glacial trough that was carved by a valley glacier extending below sea level, and subsequently drowned by marine waters

following deglaciation and post-glacial eustatic sea-level rise (common in Norway, Greenland, Alaska, and New Zealand).

3. Systematic Comparison of Alpine Glacial Erosional Landforms

Glacial Feature	Morphological Geometry	Genesis / Mechanism	Match Status in Question
Nunatak	Isolated rocky island / peak above ice sheet	Unglaciated bedrock surrounded by flowing ice	Correctly matched
Cirque	Steep-walled semicircular amphitheater basin	Headwall plucking and rotational ice scooping	Correctly matched
Arête	**Narrow, sharp-edged, knife-like ridge**	Intersection of two adjacent cirques or valleys	**True definition of the description in (c)**
Col	**Low saddle-shaped pass / gap in ridge**	Breach of an arête by two opposing cirques	**MISMATCHED in (c)**
Fjord	Drowned U-shaped glacial trough below sea level	Deep glacial valley gouging flooded by marine waters	Correctly matched

Textbook Authority

Arthur L. Bloom, *Geomorphology: A Systematic Analysis of Late Cenozoic Landforms*:
"An arête is a narrow, serrate, knife-edge ridge separating two glacial troughs or cirques. A col is a low point or saddle on an arête created by the intersection of the headwalls of two opposing cirques."
P. K. Mukherjee, *Textbook of Geology*:
"Arêtes are sharp-edged, saw-tooth ridges separating two valley glaciers. Where two cirques eat backwards towards each other, they make a breach in the dividing arête, forming a saddle or pass known as a col."